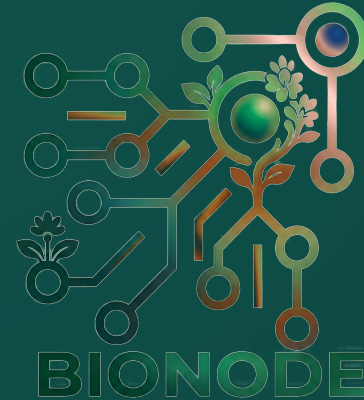




BioNode

The Omnivorous BioCNG Micro-Refinery

Bangladesh does not lack fuel. It lacks the wit to see fuel in what it throws away. The river chokes, the market rots, the field burns — **three sorrows, one answer.**

Every biogas plant before ours was built for one waste, and died when that waste stopped. Ours eats everything.

Six wastes. One appetite.

We are not a hyacinth company. We are a machine that is indifferent to its diet — and that indifference is the moat.

 Water Hyacinth Rivers, canals, haors 25–30 m³/t Free · clears waterways	 Rice Straw Boro & Aman harvest 200–300 m³/t Near-free · else burned	 Banana Stem Post-harvest pseudostem 50–70 m³/t Free · farm residue	 Kacha Bazar Waste Urban wet markets, daily 80–120 m³/t Paid to remove it	 Poultry & Dairy Litter, dung — contracted 25–120 m³/t Paid to remove it	 Agro-Mill Waste Press mud, pulp, oil cake 60–150 m³/t Tipping fee revenue
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The calendar never has a gap. Three sources run all twelve months; the seasonal ones stack on top.

	J	F	M	A	M	J	J	A	S	O	N	D
Kacha bazar waste	■	■	■	■	■	■	■	■	■	■	■	■
Poultry & dairy	■	■	■	■	■	■	■	■	■	■	■	■
Banana pseudostem	■	■	■	■	■	■	■	■	■	■	■	■
Rice straw (storable)	■	■	■	■	■	■	■	■	■	■	■	■
Water hyacinth	■	■	■	■	■	■	■	■	■	■	■	■
Agro-mill waste	■	■	■	■	■	■	■	■	■	■	■	■

No single waste makes good gas. The blend does — and the blend is always available.

How it works. Blend first, then burn nothing.



The ledger. Three revenues, one machine.

Modelled for Node-1 at 5 tonnes of blended feedstock per day. Design targets — not yet measured on a built unit.

REVENUE 1 — FUEL Tk 33 L ~200 m³/day of 200-bar BioCNG sold at the pump, per year	REVENUE 2 — TIPPING Tk 9 L Markets and mills pay us to take the waste away	REVENUE 3 — FERTILISER Tk 15 L Digestate dried and bagged as organic soil amendment	PAYBACK ~2.7 yr On ~Tk 80 lakh CAPEX, at ~Tk 30 lakh modelled EBITDA
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The gas is nearly free.

Tipping fees and fertiliser cover roughly 90% of operating cost before a single cubic metre is sold. Our competitors buy their feedstock. We are paid to accept ours.

Why this cannot be casually copied.



The chemist is the software

BioNode OS reads pH, VFA, temperature and gas quality, then re-tunes the blend recipe in real time. Anyone can weld a tank; nobody else can run one on a diet that changes weekly.



The recipe library compounds

Every tonne digested teaches the model a better blend. Hardware can be bought in a week; ten thousand hours of blend telemetry cannot.



Waste contracts are exclusive

A market or mill signs one removal contract, not five. Each node locks its catchment — the supply moat closes behind us.



Sited where the waste already is

The node stands at the market gate and the terminal forecourt. No pipeline, no import, no delivery truck — the supply chain is a wheelbarrow.

Honest about the machine. Certain about the plan.

DESIGNED — TODAY

Engineering & AI complete

Full AIoT architecture, blend model, costed BOM, six feedstock supply routes mapped.

SUMMIT · JUL 26

Bench co-digestion

Multi-feedstock 50-L reactor with live control telemetry — the diet thesis, demonstrated.

MVP · BY NOV 26

Full chain to a cylinder

Blend yard → digester → upgrade → 200 bar. First costed, deployable node.

PILOT · 2027

Three Dhaka nodes + CES

Gabtoli, Sayedabad, Mohakhali — audited field economics, real contracts.

We are not selling gas. We are building the operating system for Bangladesh's waste.



SCAN — LIVE AI DASHBOARD

Ask us the hard question: “What happens when the hyacinth dies?”

Live Edge-AI control dashboard · bionode.com · bionodebd@gmail.com

BioNode

Built in Bangladesh. Fuel for the World.

Team BioNode · IIUC · 25–26 July · Novo Theatre, Dhaka